CONTEXT

Come si è passati da GPU finalizzate al rendering 3D finalizzato al gaming fino a GPU che integrano componenti specifiche per l'AI come i tensor core.
L'hardware e il software (tramite CUDA ad esempio) hanno seguito le esigenze e il loro cambiamento nel tempo.

ORIGIN OF THE FIRST GPU (GeForce 256)

The first GPUs only performed calculations on fragments (rasterization, texture mapping z-buffering (hidden surfaces removal), shading)

While the conversion from 3D model to 2D image was performed by CPU

GEFORCE 256: introduced the first chip integrating the operation on vertices offloading the CPU

PROGRAMMABILITY AND HPC

To allow more complex operations fixed architechture were replaced by programmable shaders

Scientist adapted their math operations to "fit" with the 3D rendering operation and use the power of these GPU.

G80 (TESLA): introduced an architecture → Streaming Multiprocessors → capable of executing calculation threads independent of the graphic pipeline

introduced CUDA → new programming language

FERMI ARCHITECTURE: Added specific features for HPC: double precision computing (FP64) and memory error protection and correction

↓ reliable chip were necessary for not ruin days of calculations

(extra bit storage used to detect error and trigger algorithm for correction)

DEEP LEARNING REVOLUTIONS AND TENSOR CORE

GPU have immense floating point power and allowed the training of more complex model faster than CPU.

While HPC need high precision deep learning need high troughput at low precision

→ Forced to change moore law with Huang's law → Performance of GPU for AI

Law of transistor

Specialized hardware → Tensor core: can process massive block of matrix operation in a single hardware instruction

ORIZONTAL SCALABILITY

the increasing complexity of the model increased the computation power needed and save efficient way